VŠB TECHNICKÁ UNIVERZITA OSTRAVA | FAKULTA ELEKTROTECHNIKY A INFORMATIKY

Cooperative Localization of Two Drones Using a Kalman Filter

Semester Project

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Chapter 1 Introduction

Reliable localization is one of the basic requirements for autonomous drones. In an ideal deterministic model, the position and velocity of each drone would be known exactly. In practice, however, the true position of all the drones are not known and only imperfect sensor measurements are available. GPS measurements are especially useful because they provide absolute position information, but they can also contain significant noise. Therefore, the localization problem is naturally formulated as a state estimation problem under uncertainty.

This project studies a simplified cooperative localization problem for two drones moving in a two-dimensional plane. Each drone receives a noisy GPS measurement of its own position. In addition, the drones are assumed to obtain a noisy relative position measurement, which describes the position of Drone 2 with respect to Drone 1 (and vice versa). The main idea is that GPS provides weak absolute information, while the relative measurement provides additional information about the geometry of the two-drone formation.

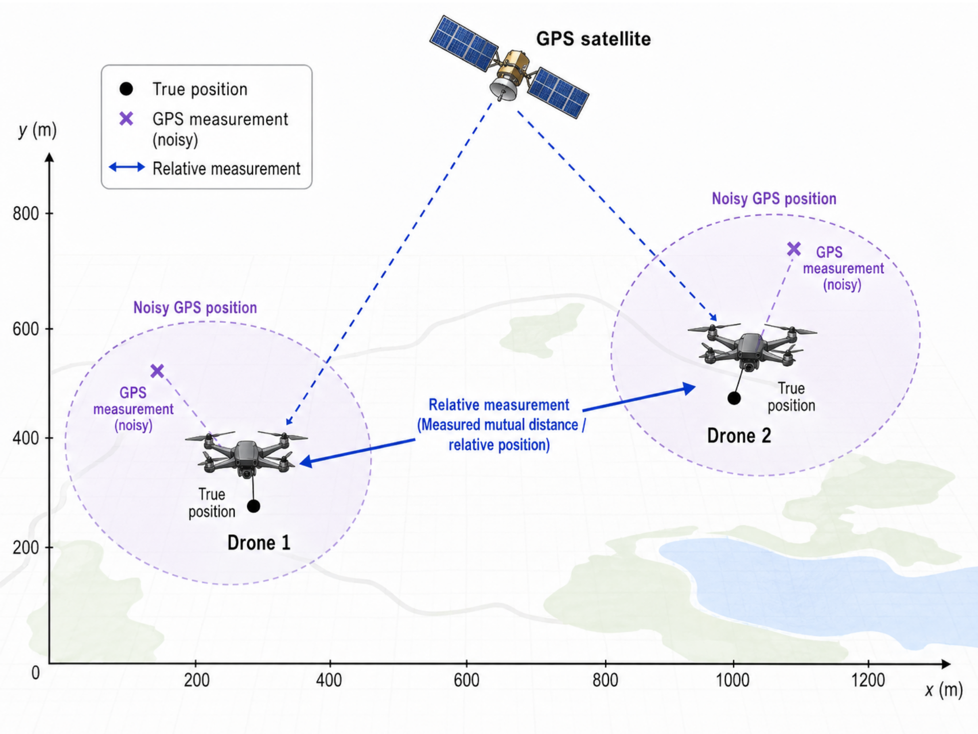

Figure 1.1: Simplified two-drone localization problem with noisy GPS measurements and an additional relative position measurement.

The hidden state of the system contains the positions and velocities of both drones. The GPS sensor measures only the absolute positions, while the relative sensor measures the vector

$$\begin{pmatrix} x_{2,k} - x_{1,k} \\ y_{2,k} - y_{1,k} \end{pmatrix}.$$

This relative measurement is intentionally chosen in this simplified form because it leads to a linear observation model and can therefore be handled by the classical linear Kalman filter. A more realistic distance-only measurement would have the form

$$d_k = \sqrt{(x_{2,k} - x_{1,k})^2 + (y_{2,k} - y_{1,k})^2},$$

which is nonlinear and would require an Extended Kalman Filter.

The goal of the project is to implement and test a Kalman filter for this two-drone localization problem. Two estimation approaches are compared: a GPS-only Kalman filter and a Kalman filter using both GPS and relative position measurements. The expected result is that the additional relative measurement improves the consistency of the estimated two-drone formation and, through the joint Kalman update, can also improve the absolute position estimates of both drones. GPS remains necessary because the relative measurement alone cannot anchor the system in the global coordinate frame.

Chapter 2 Kalman Filter Theory

The Kalman filter is a recursive algorithm for estimating the hidden state of a dynamical system from noisy measurements, which can be understood as a repeated Bayesian estimation procedure. At each time step, two operations are performed. First, the previous state estimate is propagated through the motion model. This gives a prediction of the current state before the new measurement is used. Second, this prediction is corrected using the new measurement. Conceptually, the prediction step corresponds to marginalization over the previous uncertain state, while the update step corresponds to conditioning on the newly observed measurement. In density notation, the basic Bayesian idea can be written as

$$f(x, y) = f(y \mid x)f(x) = f(x \mid y)f(y). \tag{2.1}$$

In the filtering context, $f(x)$ represents prior or predicted knowledge about the hidden state, $f(y \mid x)$ represents the measurement model, and $f(x \mid y)$ represents the posterior knowledge after observing the measurement.

The classical Kalman filter assumes a linear state-space model

$$X_k = A_k X_{k-1} + B_k u_k + W_k, \tag{2.2}$$

and a linear measurement model

$$Y_k = H_k X_k + V_k. \tag{2.3}$$

Here X_k is the hidden state at time k, Y_k is the measurement vector, u_k is a known control input, A_k is the state transition matrix, B_k is the control matrix and H_k is the measurement matrix. The random vector W_k is the process noise, which models imperfections in the motion model, and V_k is the measurement noise, which models sensor errors.

The standard assumptions are

$$W_k \sim \mathcal{N}(0, Q_k), \qquad V_k \sim \mathcal{N}(0, R_k), \tag{2.4}$$

where Q_k is the process noise covariance matrix and R_k is the measurement noise covariance matrix. In this project no explicit control input is used, so the term $B_k u_k$ is omitted in the implemented model.

The Gaussian assumption is important because Gaussian random vectors are closed under linear transformations and conditioning. This means that if the previous state estimate is Gaussian and the system model is linear with Gaussian noise, then the predicted state is again Gaussian. Similarly, if the predicted state and the measurement model are linear Gaussian, then the updated posterior state is also Gaussian. Therefore, the filter does not need to store a complete probability density. It is enough to recursively update only the mean vector and covariance matrix.

Assume that after processing measurements up to time $k - 1$, the state estimate is represented by the mean μ_{k-1} and covariance P_{k-1}. The prediction step computes the state estimate before using the current measurement:

$$\mu_k^- = A_k \mu_{k-1} + B_k u_k, \tag{2.5}$$

$$P_k^- = A_k P_{k-1} A_k^T + Q_k. \tag{2.6}$$

The vector μ_k^- is the predicted state mean and P_k^- is the predicted covariance matrix. The first equation propagates the previous mean through the deterministic part of the model. The second equation propagates the previous uncertainty and adds the process noise covariance Q_k.

Before applying the correction, the filter predicts what the sensor should measure:

$$\hat{y}_k = H_k \mu_k^-. \tag{2.7}$$

The difference between the actual measurement y_k and the predicted measurement $\hat{y}_k$ is called the innovation or residual:

$$r_k = y_k - H_k \mu_k^-. \tag{2.8}$$

The innovation represents the new information brought by the current measurement. If the innovation is small, the measurement agrees with the prediction. If it is large, the prediction and measurement disagree.

The Kalman gain determines how strongly the prediction should be corrected by the measurement:

$$K_k = P_k^- H_k^T \left(H_k P_k^- H_k^T + R_k \right)^{-1}. \tag{2.9}$$

This matrix balances two sources of uncertainty. If the predicted covariance P_k^- is large, the filter trusts the measurement more. If the measurement noise covariance R_k is large, the filter trusts the measurement less. Thus, the Kalman gain expresses the compromise between the model prediction and the sensor data.

After computing the innovation and Kalman gain, the predicted state is corrected:

$$\mu_k = \mu_k^- + K_k \left(y_k - H_k \mu_k^- \right), \tag{2.10}$$

$$P_k = P_k^- - K_k H_k P_k^-. \tag{2.11}$$

The updated mean μ_k is obtained by adding a correction term to the predicted mean. The updated covariance P_k is usually smaller than the predicted covariance, because the measurement provides additional information about the state. The pair (μ_k, P_k) then becomes the starting point for the next time step.

In this project, the same Kalman filter equations are used for two different measurement models. In the GPS-only case, the measurement matrix observes only the absolute positions of both drones. In the combined case, the measurement vector also contains the relative position components $x_{2,k} - x_{1,k}$ and $y_{2,k} - y_{1,k}$. The algorithm itself remains unchanged; only the measurement vector Y_k, the matrix H_k, and the measurement covariance matrix R_k are modified. This makes the two-drone problem a suitable example of how the Kalman filter combines different sources of noisy information in a single state estimate.

Chapter 3 Proposed Two-Drone Model

This chapter defines the mathematical model used in the numerical experiment. The goal is to formulate the two-drone localization problem in the form required by the linear Kalman filter. Therefore, the model must specify the hidden state vector, the state transition equation, the process noise covariance, the measurement vectors, the measurement matrices and the measurement noise covariances.

3.1 Modeling assumptions

The project considers two drones moving in a two-dimensional plane. The time is discrete and is denoted by

$$k = 0, 1, 2, \ldots, N.$$

The sampling period between two consecutive time steps is denoted by Δt. Both drones are assumed to follow an approximately constant-velocity model. This means that during one time step their positions are predicted from their previous positions and velocities. The velocities are not assumed to be perfectly constant in reality, but deviations from the ideal model are represented by process noise.

The drones are observed by two types of measurements. First, each drone receives a noisy GPS measurement of its absolute position in the global coordinate system. Second, the drones are assumed to obtain a noisy relative position measurement, which gives the relative displacement of Drone 2 with respect to Drone 1 (and vice versa). No explicit control input is used in this simplified model (i.e. for example no acceleration). Therefore, the general control term $B_k u_k$ from the Kalman filter theory is omitted.

All process and measurement noises are modeled as zero-mean Gaussian random vectors. This choice keeps the whole model in the linear Gaussian framework and allows the use of the classical Kalman filter.

3.2 Hidden state vector

The hidden state contains the positions and velocities of both drones:

$$X_k = \begin{pmatrix} x_{1,k} & y_{1,k} & v_{x1,k} & v_{y1,k} & x_{2,k} & y_{2,k} & v_{x2,k} & v_{y2,k} \end{pmatrix}^T . \tag{3.1}$$

Here $x_{i,k}$ and $y_{i,k}$ are the coordinates of Drone i at time k, while $v_{xi,k}$ and $v_{yi,k}$ are its velocity components in the x- and y-directions. The positions are measured by GPS, but the velocities are not measured directly. They are estimated indirectly through the motion model and the sequence of position measurements.

3.3 State transition model

For one drone with state vector

$$X_{i,k} = \begin{pmatrix} x_{i,k} & y_{i,k} & v_{xi,k} & v_{yi,k} \end{pmatrix}^T ,$$

the constant-velocity transition model is

$$X_{i,k} = F X_{i,k-1} + W_{i,k}, \tag{3.2}$$

where

$$F = \begin{pmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{3.3}$$

The first two rows express the equations

$$x_{i,k} = x_{i,k-1} + \Delta t\, v_{xi,k-1}, \qquad y_{i,k} = y_{i,k-1} + \Delta t\, v_{yi,k-1},$$

while the last two rows express the assumption that the velocity is approximately constant.

For two drones, the full state transition matrix is block diagonal:

$$A = \begin{pmatrix} F & 0 \\ 0 & F \end{pmatrix} . \tag{3.4}$$

The full state equation is therefore

$$X_k = A X_{k-1} + W_k, \tag{3.5}$$

where W_k is the process noise of the complete two-drone system.

3.4 Process noise covariance

The process noise represents the fact that the real drone motion does not follow the constant-velocity model perfectly. A convenient way to construct the covariance matrix is to assume that during one time step the drone can be affected by an unknown random acceleration.

For one spatial coordinate, for example the x-coordinate, assume that the acceleration noise is a_x. If this acceleration is approximately constant during one time step, then the position and velocity errors caused by this acceleration are

$$\Delta x = \frac{1}{2}\Delta t^2 a_x, \qquad \Delta v_x = \Delta t\, a_x.$$

Therefore, the process noise vector for one coordinate can be written as

$$w_x = \begin{pmatrix} \frac{1}{2}\Delta t^2 \\ \Delta t \end{pmatrix} a_x.$$

If

$$a_x \sim \mathcal{N}(0, \sigma_a^2),$$

then the covariance matrix of this two-dimensional noise vector is

$$Q_x = \sigma_a^2 \begin{pmatrix} \frac{1}{2}\Delta t^2 \\ \Delta t \end{pmatrix} \begin{pmatrix} \frac{1}{2}\Delta t^2 & \Delta t \end{pmatrix} = \sigma_a^2 \begin{pmatrix} \Delta t^4/4 & \Delta t^3/2 \\ \Delta t^3/2 & \Delta t^2 \end{pmatrix}. \tag{3.6}$$

The term $\Delta t^4/4$ is the variance contribution to the position, the term Δt^2 is the variance contribution to the velocity, and the term $\Delta t^3/2$ is the covariance between the position and velocity errors caused by the same unknown acceleration.

For one drone moving in two dimensions, the same construction is used independently in the x- and y-directions. With the state ordering

$$(x,\ y,\ v_x,\ v_y)^T,$$

the process noise covariance of one drone is

$$Q_{\text{drone}} = \sigma_a^2 \begin{pmatrix} \Delta t^4/4 & 0 & \Delta t^3/2 & 0 \\ 0 & \Delta t^4/4 & 0 & \Delta t^3/2 \\ \Delta t^3/2 & 0 & \Delta t^2 & 0 \\ 0 & \Delta t^3/2 & 0 & \Delta t^2 \end{pmatrix}. \tag{3.7}$$

The parameter σ_a controls how strongly the real motion is allowed to deviate from the ideal constant-velocity model. A larger value means that the filter trusts the motion model less.

For the complete two-drone system, the process noise covariance matrix is

$$Q = \begin{pmatrix} Q_{\text{drone}} & 0 \\ 0 & Q_{\text{drone}} \end{pmatrix}. \tag{3.8}$$

This assumes that the process noises of the two drones are independent.

3.5 GPS-only measurement model

The first estimation approach uses only GPS measurements. The GPS measurement vector contains the noisy absolute positions of both drones:

$$Y_k^{GPS} = \begin{pmatrix} x_{1,k}^{GPS} \\ y_{1,k}^{GPS} \\ x_{2,k}^{GPS} \\ y_{2,k}^{GPS} \end{pmatrix}. \tag{3.9}$$

This can be written in the linear measurement form

$$Y_k^{GPS} = H_{GPS} X_k + V_k^{GPS}, \tag{3.10}$$

where

$$H_{GPS} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}. \tag{3.11}$$

The first two rows select the position coordinates of Drone 1, and the last two rows select the position coordinates of Drone 2. The velocities are not measured directly.

The GPS measurement noise is modeled as

$$V_k^{GPS} \sim \mathcal{N}(0, R_{GPS}), \qquad R_{GPS} = \sigma_{GPS}^2 I_4. \tag{3.12}$$

The parameter σ_{GPS} represents the standard deviation of the GPS position error.

3.6 GPS and relative position measurement model

The second estimation approach uses both GPS and relative position measurements. The relative measurement is defined as

$$Y_k^{rel} = \begin{pmatrix} x_{2,k} - x_{1,k} \\ y_{2,k} - y_{1,k} \end{pmatrix} + V_k^{rel}. \tag{3.13}$$

It describes the position of Drone 2 with respect to Drone 1. This is not an absolute position measurement. It only constrains the geometry of the two-drone formation.

The complete measurement vector for the combined model is

$$Y_k^{comb} = \begin{pmatrix} x_{1,k}^{GPS} \\ y_{1,k}^{GPS} \\ x_{2,k}^{GPS} \\ y_{2,k}^{GPS} \\ x_{2,k} - x_{1,k} \\ y_{2,k} - y_{1,k} \end{pmatrix} + V_k^{comb}. \tag{3.14}$$

This can again be written as a linear measurement model

$$Y_k^{comb} = H_{comb} X_k + V_k^{comb}, \tag{3.15}$$

where

$$H_{comb} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}. \tag{3.16}$$

The first four rows are identical to the GPS-only measurement matrix. The fifth row computes $x_{2,k} - x_{1,k}$, and the sixth row computes $y_{2,k} - y_{1,k}$.

The measurement noise covariance matrix for the combined model is

$$R_{comb} = \text{diag}\left(\sigma_{GPS}^2, \sigma_{GPS}^2, \sigma_{GPS}^2, \sigma_{GPS}^2, \sigma_{rel}^2, \sigma_{rel}^2\right), \tag{3.17}$$

where σ_{rel} is the standard deviation of the relative position measurement error. If σ_{rel} is smaller than σ_{GPS}, then the relative sensor provides more accurate information about the formation geometry than GPS alone.

3.7 Comparison of the two estimation models

The two filters compared in this project use the same hidden state vector, the same transition matrix A, and the same process noise covariance matrix Q. The only difference is in the measurement model. The GPS-only filter uses Y_k^{GPS}, H_{GPS}, and R_{GPS}. The combined filter uses Y_k^{comb}, H_{comb}, and R_{comb}.

This makes the comparison fair, because both filters use the same motion model. Any difference in the results is caused by the additional relative position measurement. The expected effect is that the relative measurement improves the consistency of the estimated two-drone formation. Since the Kalman update is applied to the whole state vector, this relative information can also improve the absolute position estimates of both drones when it is combined with GPS. However, the relative measurement alone cannot determine the global position of the whole system. GPS remains necessary to anchor the formation in the global coordinate frame.

Chapter 4 Implementation

The numerical implementation is written in Python. The library `numpy` is used for vector and matrix computations, while `matplotlib` is used for plotting the resulting trajectories, errors and RMSE comparison. The animation of the localization process is created using `matplotlib.animation`. The implementation is intentionally procedural and compact.

The program first creates all matrices required by the two compared filters, namely the transition matrix A, process noise covariance Q, GPS measurement matrix H_{GPS}, combined measurement matrix H_{comb}, and corresponding measurement noise covariance matrices R_{GPS} and R_{comb}. Then it simulates the true hidden trajectory of both drones, generates noisy GPS and relative position measurements, and runs two Kalman filters. The first filter uses only GPS measurements, while the second filter uses both GPS and the relative position measurement. After filtering, the code computes absolute position errors, relative position errors and RMSE values. Finally, it creates graphical outputs showing the true trajectories, noisy measurements, Kalman estimates, error curves, RMSE comparison and an animation of the estimation process.

In addition to the main two-drone implementation, an early working version of a three-drone model was also prepared. This version should not be understood as a definitive part of the project, but rather as an implementation exercise showing that the state-space formulation can be extended to a larger number of drones. In the three-drone version, which uses the same Kalman filter, the state vector is enlarged to contain the positions and velocities of three drones, and the combined measurement model uses pairwise relative position measurements between the drones. The main project, however, remains focused on the simplest two-drone case, because this case is sufficient to demonstrate the main Kalman filter principle clearly.

Chapter 5 Experiments and Results

The implementation was tested on two simulated two-drone scenarios. The first set represents a favorable case with smooth motion and accurate relative position measurement. It was defined by the parameters $\Delta t = 1.0$, $N = 100$, $\sigma_a = 0.015$, $\sigma_{GPS} = 6.0$, and $\sigma_{rel} = 0.5$. The second set represents a more difficult case with stronger deviations from the ideal constant-velocity model and noisier measurements. It was defined by the parameters $\Delta t = 1.0$, $N = 100$, $\sigma_a = 0.20$, $\sigma_{GPS} = 10.0$, and $\sigma_{rel} = 3.0$. In both cases the same two filters were compared: the GPS-only Kalman filter and the Kalman filter using both GPS and the relative position measurement. The three-drone version of the code was prepared only as an additional extension and was not experimentally evaluated here. The three-drone version of the code was prepared only as an additional extension and was not experimentally evaluated here.

The absolute error represents the mean Euclidean distance between the true and estimated absolute positions of both drones. Let

$$p_{1,k} = \begin{pmatrix} x_{1,k} \\ y_{1,k} \end{pmatrix}, \qquad p_{2,k} = \begin{pmatrix} x_{2,k} \\ y_{2,k} \end{pmatrix}$$

be the true positions of Drone 1 and Drone 2 at time step k, and let $\hat{p}_{1,k}$ and $\hat{p}_{2,k}$ be their estimated positions obtained from the Kalman filter. The absolute position errors of the individual drones are

$$e_{1,k}^{abs} = \left\| \hat{p}_{1,k} - p_{1,k} \right\|_2, \qquad e_{2,k}^{abs} = \left\| \hat{p}_{2,k} - p_{2,k} \right\|_2,$$

and the plotted mean absolute error is

$$e_k^{abs} = \frac{e_{1,k}^{abs} + e_{2,k}^{abs}}{2}.$$

This quantity describes how accurately the filter estimates the global positions of the drones in the x, y coordinate system. The results in figure 5.1 show that the combined filter gives a lower absolute position error in both tested cases. The improvement is more visible in the first set, where the relative measurement is more accurate.

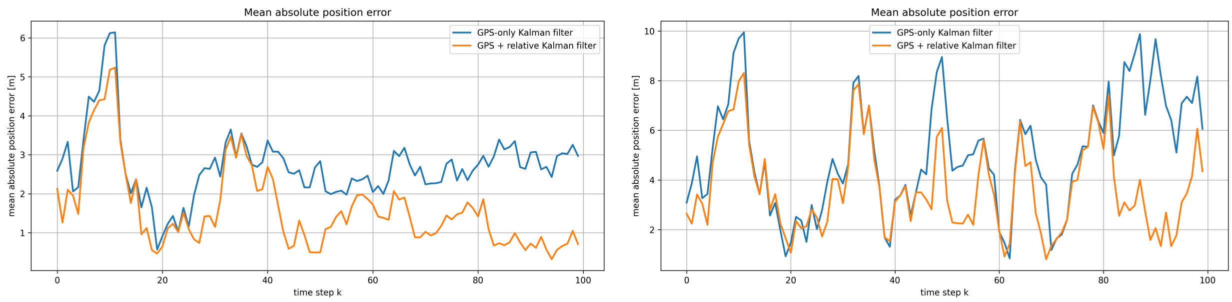

Figure 5.1: Mean absolute position error for Set 1 and Set 2.

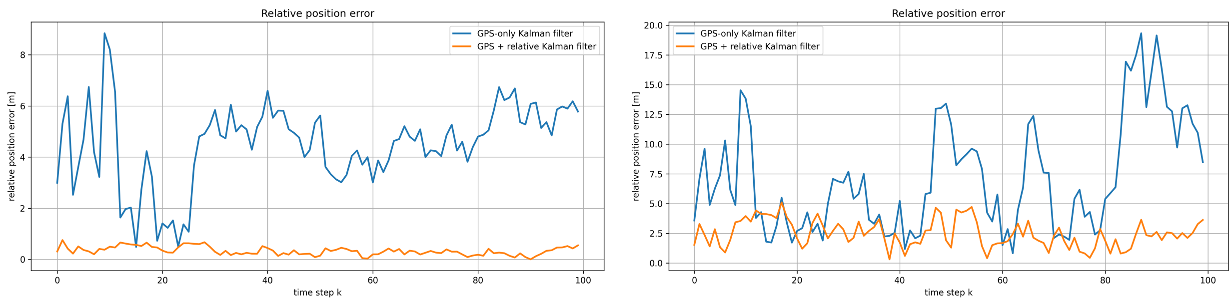

Figure 5.2: Relative position error for Set 1 and Set 2.

The relative error evaluates the accuracy of the estimated relative position of the two-drone formation. The true relative position vector is

$$r_k = p_{2,k} - p_{1,k} = \begin{pmatrix} x_{2,k} - x_{1,k} \\ y_{2,k} - y_{1,k} \end{pmatrix},$$

while the corresponding estimated relative position vector is

$$\hat{r}_k = \hat{p}_{2,k} - \hat{p}_{1,k} = \begin{pmatrix} \hat{x}_{2,k} - \hat{x}_{1,k} \\ \hat{y}_{2,k} - \hat{y}_{1,k} \end{pmatrix}.$$

The relative position error is then defined as

$$e_k^{rel} = \|\hat{r}_k - r_k\|_2 .$$

Written component-wise, this is

$$e_k^{rel} = \sqrt{[(\hat{x}_{2,k} - \hat{x}_{1,k}) - (x_{2,k} - x_{1,k})]^2 + [(\hat{y}_{2,k} - \hat{y}_{1,k}) - (y_{2,k} - y_{1,k})]^2}.$$

This quantity describes how accurately the filter estimates the internal geometry of the two-drone formation. It is the quantity that should be most directly improved by the additional relative measurement, because the combined measurement model explicitly observes $x_{2,k} - x_{1,k}$ and $y_{2,k} - y_{1,k}$. The results in figure 5.2 confirm this expectation. In both sets the combined filter has significantly lower relative error than the GPS-only filter.

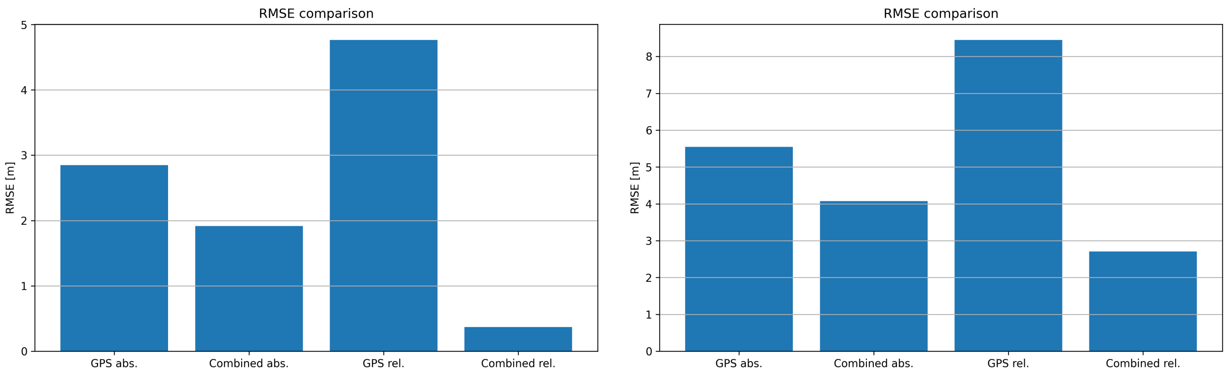

Figure 5.3: RMSE comparison for Set 1 and Set 2.

Finally, the root mean square error (RMSE) comparison in figure 5.3 summarizes each complete error curve into one scalar value. For a general error sequence $e_0, e_1, \ldots, e_{N-1}$, the RMSE is defined as

$$RMSE(e) = \sqrt{\frac{1}{N} \sum_{k=0}^{N-1} e_k^2}.$$

In this project this formula is applied separately to the absolute error sequence e_k^{abs} and to the relative error sequence e_k^{rel}. Therefore,

$$RMSE_{abs} = \sqrt{\frac{1}{N} \sum_{k=0}^{N-1} (e_k^{abs})^2}, \qquad RMSE_{rel} = \sqrt{\frac{1}{N} \sum_{k=0}^{N-1} (e_k^{rel})^2}.$$

The RMSE values are useful because they summarize the whole time-dependent behavior of the filter into a compact numerical comparison. The important observation is that the combined filter gives lower RMSE values for the relative position error in both scenarios, and also improves the absolute position error in these tests. This agrees with the theoretical interpretation: the relative measurement directly constrains the formation geometry, and because the Kalman update is applied to the whole joint state vector, this information can also improve the absolute position estimates when combined with GPS.

Chapter 6 Limitations and Practical Context

The model used in this project is intentionally simplified. The most important simplification is the form of the relative measurement. In the implemented model, the relative sensor directly provides the vector

$$\begin{pmatrix} x_{2,k} - x_{1,k} \\ y_{2,k} - y_{1,k} \end{pmatrix}.$$

This is useful for the project because it gives a linear measurement equation and therefore allows the use of the classical linear Kalman filter. In practical drone systems, however, relative measurements are usually not available in this exact form. Sensors such as ultra-wideband radios, cameras, LiDAR, radar or combinations with inertial units usually provide range, bearing, visual features or relative pose information. These measurements are often nonlinear functions of the drone positions. For example, a distance-only measurement depends on

$$d_k = \sqrt{(x_{2,k} - x_{1,k})^2 + (y_{2,k} - y_{1,k})^2}.$$

Such a measurement would no longer fit directly into the linear Kalman filter model and would usually require an Extended Kalman Filter, Unscented Kalman Filter, particle filter, graph optimization method or another nonlinear estimation approach.

Similar ideas are nevertheless used in research and practical cooperative localization. A common approach is to combine global measurements, such as GPS or GNSS, with relative measurements between vehicles. Ultra-wideband technology is often used for this purpose, because it can provide relatively accurate inter-agent distance measurements and can also work in environments where GPS is weak or unavailable. Other approaches combine visual, inertial and range measurements, where cameras and inertial sensors provide local motion information and range sensors constrain the relative geometry of the swarm.[1]

Compared with these practical systems, the present project should be understood as a simplified linear demonstration of the same general principle. GPS anchors the formation in the global coordinate frame, while relative measurements strongly improve the estimated internal geometry of the formation. In the performed experiments, this additional geometric information also slightly improved the absolute position estimates of both drones, because the Kalman update is applied to the whole joint state vector. Therefore, the relative measurement does not only correct the distance between the drones, but can also help the global position estimate when it is fused with GPS. The project still does not solve the full practical drone-swarm localization problem, because it does not include nonlinear range measurements, sensor delays, communication limits, obstacle visibility, outliers, or real flight dynamics. Its purpose is to implement a clear and functional Kalman filter example showing how absolute and relative information can be fused to improve both formation consistency and, in this simplified setting, also the absolute localization accuracy.

Chapter 7 Conclusion

This project demonstrated the use of a Kalman filter for a simplified cooperative localization problem involving two drones in a two-dimensional plane. The problem was formulated as a linear Gaussian state-space model, where the hidden state contained the positions and velocities of both drones. Two estimation approaches were implemented and compared. The first one used only noisy GPS measurements, while the second one combined GPS measurements with an additional relative position measurement between the drones.

The main goal of the project was to create a functional implementation of the Kalman filter and to show how different sources of uncertain information can be fused in one joint state estimate.

[1] Examples of related work include: *UWB-Based Localization for Multi-UAV Systems and Collaborative Heterogeneous Multi-Robot Systems*; *Cooperative Relative Localization in MAV Swarms with Ultra-wideband Ranging*; and *Performance Analysis of Visual–Inertial–Range Cooperative Localization for Unmanned Autonomous Vehicle Swarm.*

This was achieved by implementing the complete filtering pipeline in Python, including model construction, trajectory simulation, measurement generation, prediction and update steps, error evaluation, graphical outputs and animation. The same Kalman filter equations were used in both compared cases, while only the measurement vector, measurement matrix and measurement noise covariance matrix were changed.

The experiments confirmed the expected behavior of the model. The additional relative position measurement significantly improved the estimated internal geometry of the two-drone formation. In the tested scenarios, it also improved the absolute position estimates, because the Kalman update was applied to the whole joint state vector. GPS still remained necessary, because the relative measurement alone cannot determine the global position of the formation.

The model is intentionally simplified and does not represent a complete practical drone-swarm localization system. In real applications, relative measurements are often nonlinear and may require an Extended Kalman Filter or another nonlinear estimation method. Nevertheless, the project shows the basic principle clearly: absolute measurements can anchor the system globally, while relative measurements can improve the consistency and accuracy of the joint estimate. Therefore, the implemented model provides a useful first step toward understanding cooperative localization with Kalman filtering.

Finally, there are two natural directions in which the present model could be developed further. The first direction is to generalize the implementation from two drones to an arbitrary number of drones. For n drones, the hidden state would have the form

$$X_k = \begin{pmatrix} x_{1,k} & y_{1,k} & v_{x1,k} & v_{y1,k} & \cdots & x_{n,k} & y_{n,k} & v_{xn,k} & v_{yn,k} \end{pmatrix}^T \in \mathbb{R}^{4n}.$$

The transition matrix would remain block diagonal,

$$A_n = \operatorname{diag}(F, F, \dots, F),$$

where F is the one-drone constant-velocity transition matrix. The GPS measurement model would also be extended in a systematic way by selecting the position coordinates of all drones. Relative measurements could be added either between selected pairs of drones or between all pairs. In this sense, the method is structurally scalable, but the implementation would have to be rewritten in a more general form, where matrices are generated automatically for arbitrary n, instead of being written explicitly for a fixed number of drones.

The second natural direction is to replace the simplified relative position measurement by a more realistic distance-only measurement. Instead of measuring the vector

$$\begin{pmatrix} x_{j,k} - x_{i,k} \\ y_{j,k} - y_{i,k} \end{pmatrix},$$

a practical range sensor would more naturally measure only the Euclidean distance

$$d_{ij,k} = \sqrt{(x_{j,k} - x_{i,k})^2 + (y_{j,k} - y_{i,k})^2} + v_{ij,k}.$$

This measurement is more realistic, but it is nonlinear in the state variables. Therefore, it cannot be used directly with the classical linear Kalman filter in the form used in this project. A model based on such measurements would require an Extended Kalman Filter, where the nonlinear measurement function $h(X_k)$ is linearized around the current predicted state using its Jacobian matrix,

$$H_k = \left. \frac{\partial h}{\partial X} \right|_{X=\mu_k^-}.$$

This would be a more realistic but also significantly more complex continuation of the present work.